

EggHand: A Multimodal Foundation Model for Egocentric Hand Pose Forecasting

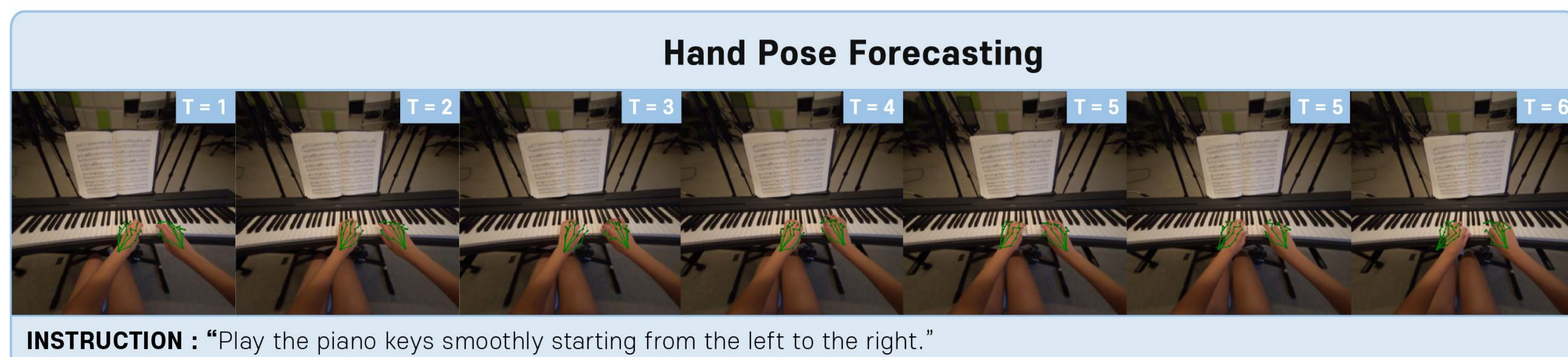
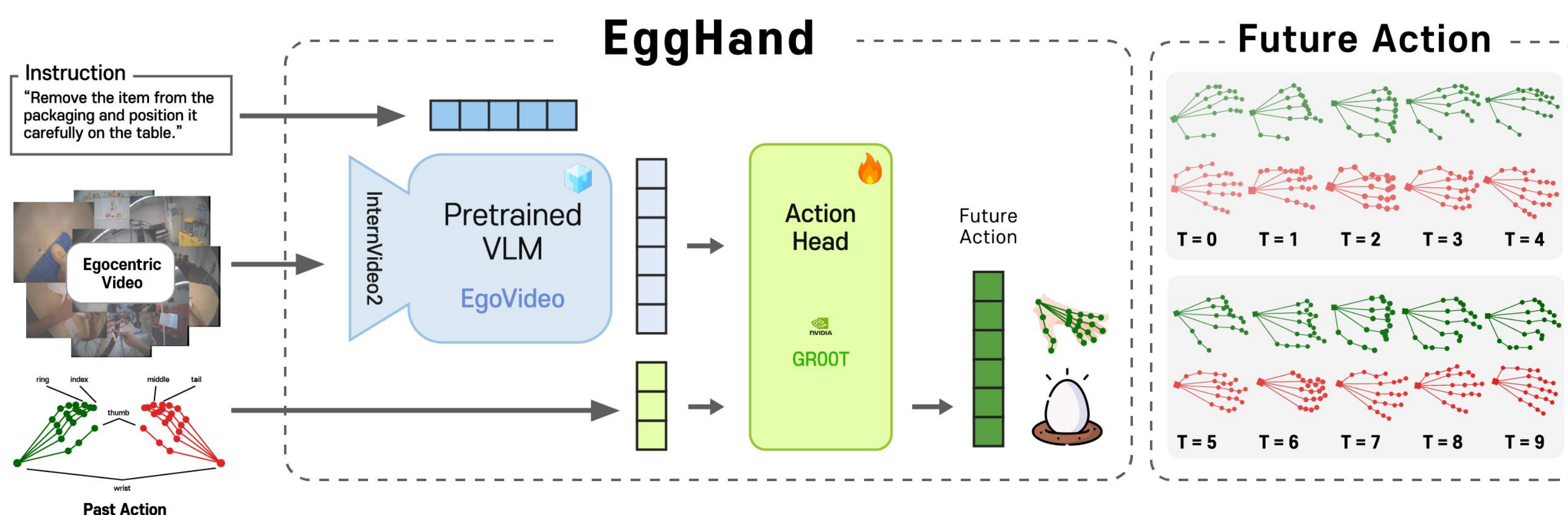
Jaeyoung Choi*, Hyeondong Kim*, Yujin Kim, Daehee Park†
¹Daegu Gyeongbuk Institute of Science & Technology (DGIST)

Introduction

- Egocentric 3D hand pose forecasting is essential for anticipating human intentions in AR/VR and human-robot interaction.
- However, existing methods rely on low-level motion cues, failing to capture high-level semantics and degrading under severe camera ego-motion.
- We propose **EggHand**, combining a VLA action decoder with an egocentric video encoder for robust, viewpoint-aware future forecasting.

Motivation

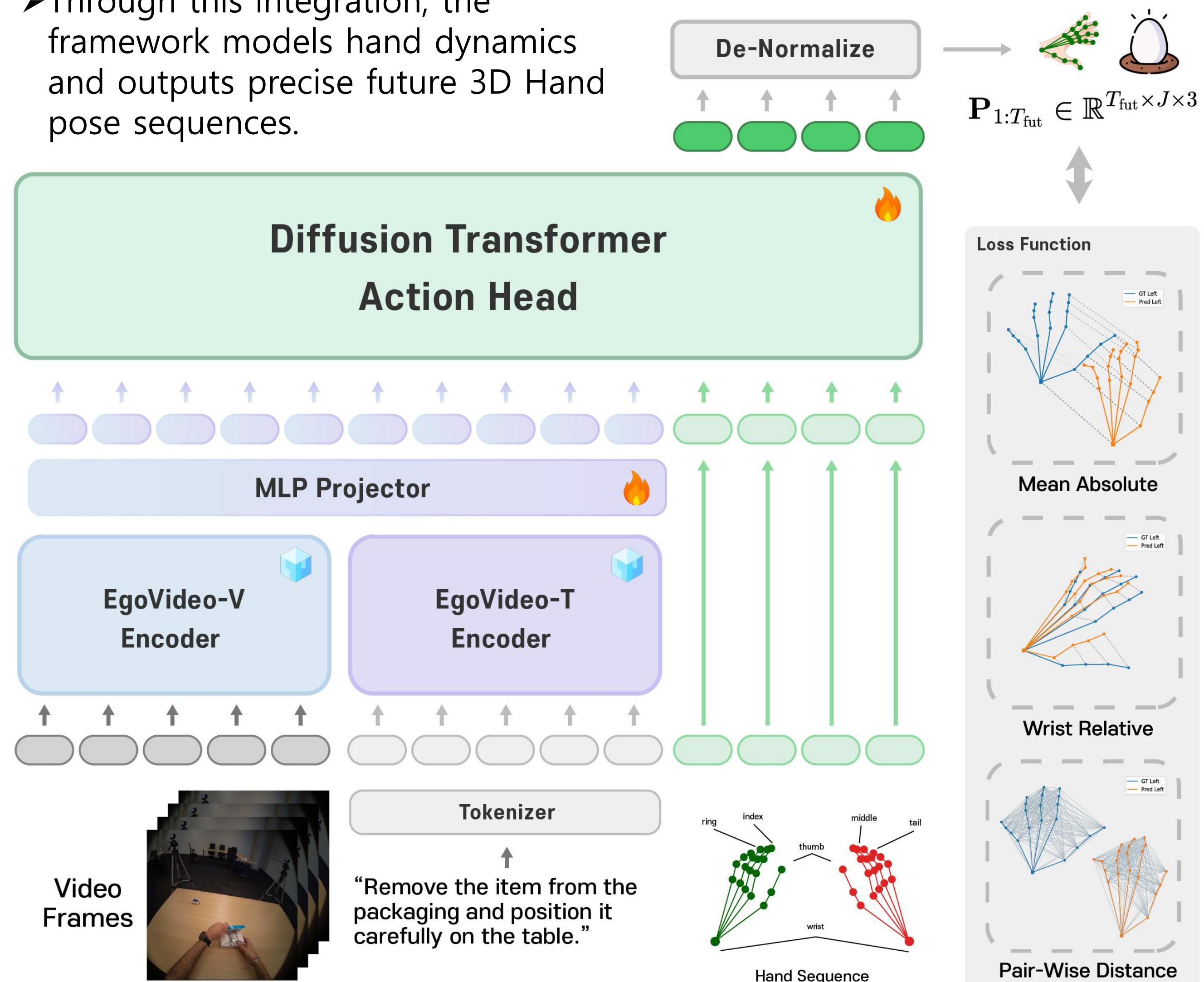
- While VLA models excel at physically meaningful action generation, they have not yet been applied to egocentric 3D hand pose forecasting.
- VLA's VLM encoders fail to capture complex, high-dimensional hand kinematics and strict joint constraints under severe camera motion.



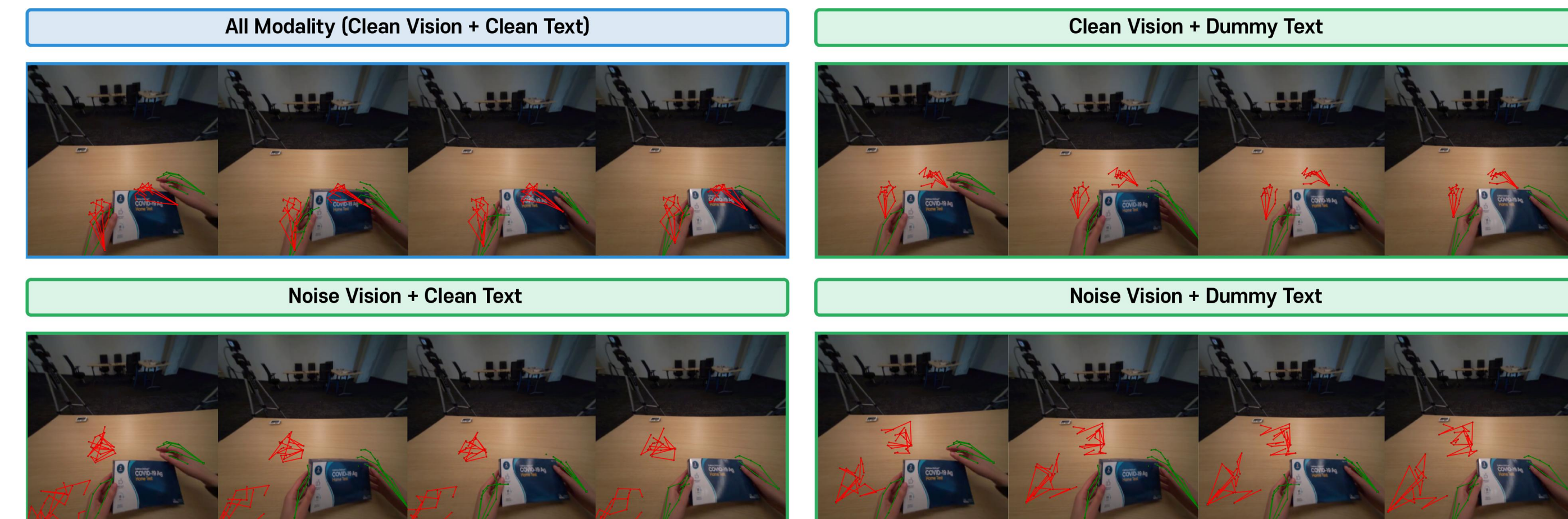
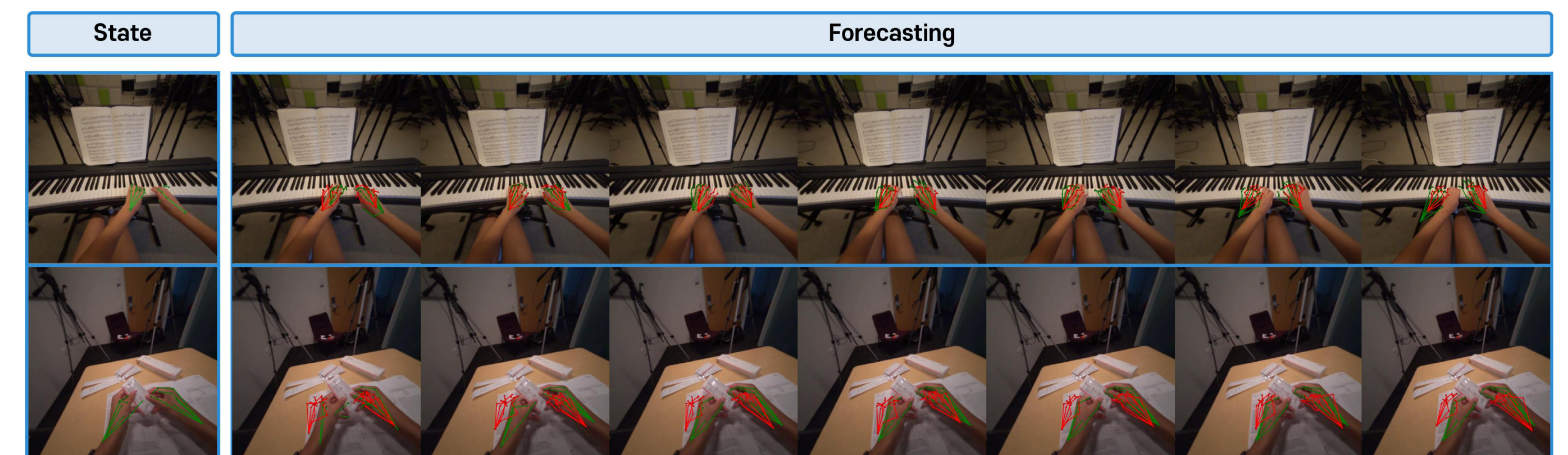
Method

Key Idea

- **Foundation Model Domain Adaptation:** Fusing an egocentric foundation encoder with a VLA decoder for stable, kinematics-aware forecasting.
- We connect a frozen egocentric video encoder (EgoVideo) with a GROOT-based VLA decoder using lightweight adapters to blend spatial context with temporal action priors.
- The network is optimized via a geometry-aware loss that jointly enforces absolute accuracy, wrist-relative stability, and intra-hand joint consistency.
- Through this integration, the framework models hand dynamics and outputs precise future 3D Hand pose sequences.



Results



Method	Trajectory		Pose	
	ADE	FDE	MPJPE	MPJPE-F
Static	0.335	0.405	0.166	0.179
CVM	0.346	0.467	0.166	0.183
USST	0.562	0.581	-	-
EgoH4	0.267	0.333	0.116	0.141
Ours	0.271	0.271	0.076	0.077

Method	Top 10%		All	
	ADE	MPJPE	ADE	MPJPE
EgoH4	0.294	0.139	0.267	0.116
Ours w. InternVL3.5-1B	0.303	0.112	0.287	0.082
Ours	0.276	0.106	0.271	0.076

Conclusion

- We presented **EggHand**, the first framework to successfully unlock VLA models for robust, semantics-aware egocentric 3D hand pose forecasting.
- Our approach sets a new state-of-the-art on EgoExo4D, bridging the gap between foundation models and fine-grained embodied AI applications.